Discovering Plate Approach Patterns Among Major League Baseball Batters Through Data Mining

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***Abstract*—This project examines patterns in Major League Baseball (MLB) batting behavior by analyzing plate approach statistics from qualified batters over the 2021-2025 seasons. The analysis uses the Baseball Savant MLB leaderboard, a publicly available dataset, and focuses on discovering natural sets of plate approaches from player-season observations based on the data in the leaderboard. This data measures plate discipline through swing and contact rates and includes observed results and expected results to view their effectiveness. By cross-comparing this data between the identified groups, it will be seen how each one relates to performance and outcome variance, focusing on pure pattern discovery.**

I. Introduction

The purpose of this project is to discover patterns in MLB batter data, revealing the different approaches and strategies players have when at the plate. This data can reveal the underlying discipline a player has, along with their traits. Are they an aggressive or passive hitter? Do they make consistent contact when they swing? These types of metrics determine where each player excels, identifying their skill and style type, no matter how successful they are, which can be useful in evaluating and comparing them on a comprehensive level.

II. Dataset Description

A. Data Source

The dataset to be used is derived from the Baseball Savant MLB leaderboard, and the data will be exported through the platform's built-in CSV download functionality. The dataset displays performance data among qualified MLB batters from the past five seasons (2021-2025), defined as players who have had at least 500 plate appearances in a single season.

B. Data Size

In the selected dataset, there are 669 rows, each representing a player-season observation, and 11 columns representing plate approach measurements and outcome statistics.

C. Data Representation

The dataset describes individual plate approach behavior and tendencies as well as outcome quality for an overall season. Plate approach statistics track how often a batter swings at a pitch, where they swing, and how frequently they make contact with the ball. These measurements relate to result statistics, summarized by weighted on-base average (wOBA), signifying the observed outcomes of each plate approach. Expected statistics (xwOBA) are included to measure underlying performance based on the quality of contact made, rather than observed results, helping to exclude outcome variance due to luck. The data only accounts for a player's overall performance across an entire season, not a game-by-game analysis.

D. Key Measurements

Observation Data

- Player
- Season

Plate Approach Data

- K% - Strikeout Rate
- BB% - Walk Rate
- Z-Swing% - In-Zone Swing Rate
- O-Swing% - Out-of-Zone Swing Rate
- Z-Contact% - In-Zone Contact Rate
- O-Contact% - Out-of-Zone Contact Rate
- Swing% - Swing Rate

Performance Data

- wOBA - Weighted On-Base Average
- xwOBA - Expected Weighted On-Base Average

E. Sample Data Table

A miniature sample data table is given below in Table 1 to provide a visualization of how the data is formatted. This sample shows a condensed form of the dataset across three observations and six attributes.

F. Data Quality Considerations

There are limited data quality issues in the dataset that will have to be addressed. The dataset is restricted to qualified batters, excluding part-time or bench players, which favors the analysis toward established, full-time players. Additionally,

TABLE I
Sample Data

Player	Year	K%	BB%	wOBA	xwOBA
Tucker, Kyle	2021	15.9	9.3	.383	.394
Olson, Matt	2025	24.3	12.6	.366	.360
Swanson, Dansby	2024	24.3	9.1	.307	.324

because all data is taken at a season level, any in-season variation of individual plate approach and performance changes is masked. The dataset also does not account for the opposing pitcher, varying in-game situational plate appearances, or stadium/field differences, which may influence plate approach and outcome statistics.

III. Discovery Questions

A. Questions

1) Are there distinct, natural groupings of MLB batters based on their plate approach statistics? : This is the key question that will be answered: finding patterns arising from how batters approach the plate and which groupings form from them.

2) How do observed result statistics (wOBA) and expected result statistics (xwOBA) differ across these groupings? : This question examines how the different, discovered plate approach profiles are related to offensive results, both in what is observed and what is expected based strictly on quality of contact.

3) Are there outlier player performances that significantly deviate from what is expected from their plate approach among the groups? : This question will help identify unusual observations that feature offensive performances that do not align with the typical results seen from their respective plate approach grouping.

B. Value

Overall, these discovery questions are valuable because they reveal broad plate approach profiles among MLB batters. Furthermore, these profiles can be related to performance outcomes and variance when analyzing observed and expected outcome metrics. Finding outliers enhances the analysis by uncovering pattern exceptions, which offer insight into which plate approaches tend to result in over- or underperformance when compared to expectations.

IV. Planned Data Mining Techniques

A. Clustering

K-Means Clustering will be employed to determine whether distinct, natural groupings of MLB batters exist based on their plate approach statistics. Clustering will directly address whether common plate approach profiles arise from the data by grouping player seasons using swing rates, contact rates, and discipline measures. Utilizing this clustering method will verify the validity of the spherical data types and ensure that the found patterns within them are concrete.

B. Anomaly Detection

Anomaly detection will be performed mainly through distance-based techniques to accomplish two goals. Firstly, data point distances from cluster centroids will be taken to locate players who drift farthest from their assigned cluster. These are edge-case hitters who do not align exactly with one plate approach but take on a hybrid strategy instead. Then, anomaly detection will be used to identify outliers in player-seasons that greatly over- or underperform their expected outcomes when compared to other batters within their plate approach profile grouping. Examining differences between wOBA and xwOBA statistics through Z-Scores within clusters will directly locate this unusual behavior, indicating extreme skill or variance in a batter, or just an unlikely occurrence of a plate approach profile deviating from typical patterns. The first anomaly detection implementation focuses on integrated plate approaches, while the second concentrates on irregular performance divergence.

V. Planned Analysis Pipeline

Fig. 1 depicts the planned data analysis pipeline of the project. Data inspection and preprocessing will consist of collecting, cleaning, and finalizing the dataset. Exploratory data analysis will be performed through detailed univariate, bivariate, and multivariate analysis. Feature engineering, selection, and transformation will finalize the features to be used in clustering and will ensure that the data is transformed in a way to properly facilitate clustering algorithms. Finally, the data mining stage will apply the techniques communicated in Section IV to the data to highlight its patterns and characteristics, whilst also drawing interpretative value from them.

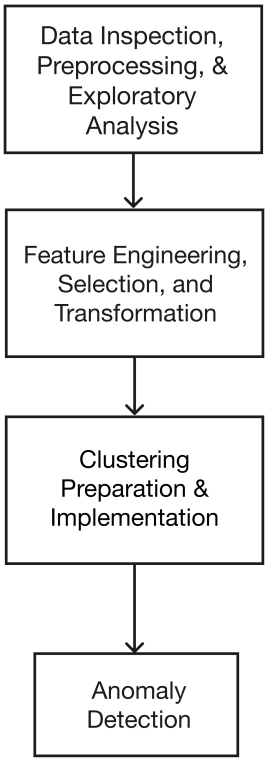

Fig. 1. Pipeline Flowchart

VI. Exploratory Data Analysis

A. Data Inspection & Preprocessing

The preparatory inspection and preprocessing of the dataset found no critical errors. Each row and column held valid values and were properly filled in, and no duplicate values were encountered in the data. The preliminary distributions of each feature appeared to be normal and standard regarding central tendency and overall dispersion.

B. Univariate Analysis

The univariate analysis of each feature yielded similar results to each other. The distributions exhibited a unimodal distribution on their respective scales, displayed through histograms and box plots. Variations included right skews, left skews, or neither, revealing several outliers on both ends of some represented graphs, but nothing much farther than that. In short, these examinations produced no prominent results, paving the way to the contrasting of features to discover the intricacies of the dataset as a whole.

C. Bivariate and Multivariate Analysis

When comparing these features through bivariate and multivariate analyses, direct and inverse relationships appeared. Strong correlations arose between the different wOBA statistics, contact percentages, swing rates, and strikeouts. A correlation heatmap revealed these patterns, and subsequent scatterplots were formed to confirm them. Together, they proved that the statistics grow in meaning when joined in groups, establishing the need for a diverse set of features to demonstrate key relationships in the dataset.

VII. Feature Engineering, Selection, & Transformation

A. Feature Engineering & Selection

The dataset features needed modifications before beginning the data mining phase. Firstly, a performance difference statistic was created to measure the difference between a player's wOBA and their xwOBA. This data point provides the essential information to view whether a player under- or overperformed in a given season, without taking into account the strength of their recorded performance outcome. This is ideal for identifying clusters with better or worse expected numbers to better determine the stronger plate approaches. Additionally, the Swing% characteristic was removed. As found in the bivariate analysis, this measurement only served as a mean between Z-Swing% and O-Swing%, making its purpose redundant. Therefore, it was omitted to clarify cluster creation and produce clearer relationship evaluations. All other features were kept as is.

B. Data Transformation

The data was transformed for clustering through standardization. Each feature was on a percentage scale, but not properly aligned regarding its range, which would break future implemented algorithms. Standardization solved this problem, bringing each range proportional to each other for proper comparisons, retaining variability and outlier handling while centering each one around zero with consistent standard deviations.

VIII. Clustering Analysis

A. Algorithm Selection

K-Means was chosen for cluster creation because the dataset was proven to be arranged normally, making it best suited to spherical cluster centers to represent precise and meaningful batter profiles.

B. Number of Clusters

The process of choosing the number of clusters to extract from the dataset was performed through the elbow method and by taking corresponding silhoutte scores. Based on the elbow method, a value of k = 3 was settled upon as the best value for maximizing in-cluster variance while minimizing total clusters. The most competitive silhouette scores came from k values 2 and 3, with the former holding a marginal edge. Despite this, a total of three clusters to pull from the dataset was kept to make for a more diverse group to broaden relationship understanding and make them richer. This decision retains knowledge without sacrificing dataset complexities, key attributes, and oversimplification, making for a more comprehensive layout to analyze in-depth.

C. Cluster Characterization

Clustering implementation found three distinct clusters. A grouped bar chart and varied scatterplots indicate these sections of clusters and visualize them distinctly.

1) Cluster 1: Cluster 1 ('Selective Batters', 196 points) holds hitters who show patience at the plate. They have the lowest swing and contact rates, but the highest walk and strikeout rates, displaying extreme plate discipline for better or worse. Their performance is also significantly the best of the clusters, and they notably underperform their expected statistics as well. In theory, this makes them the best plate approach regarding average offensive value.

2) Cluster 2: Cluster 2 ('Swing-Heavy Batters', 251 points) holds hitters who demonstrate lots of plate aggression. They hold the highest swing rates at the price of high strikeout and low walk rates. While some batters can perform very well by using this strategy, many can also suffer from it, describing the approach as a double-edged sword. Their performance falls much more in line with the mean wOBA, and they underperform that number by a very small amount.

3) Cluster 3: Cluster 3 ('Contact Batters', 222 points) holds hitters who make well-above-average contact with balls, holding the clear best numbers in this category. They also strikeout the fewest between the three clusters. Evidently, their utility comes from putting the ball in play, opening up opportunities on the field, and leaving more room for luck than usual. Performance-wise, they are on par with Cluster 2, but slightly overperform their expected statistics instead.

D. Preliminary Findings & Interpretation

The first two proposed discovery questions can now be answered. There are distinct groupings of MLB batters based on their plate approach: aggression-based, contact-based, and swing-heavy. They all feature their own unique characteristics and lead in the approach measurements for their specific specialty. Each approach exhibits its own patterns and territory, cementing cluster clarity. Observed result statistics differentiate most within Cluster 1, which marks a considerable advantage over the other two clusters. Their offensive value is generally the most effective of all. Expected result statistics are also most strongly pronounced in Cluster 1, which largely underperforms from what is expected. The other clusters only very slightly over- or underperform, making this distinction notable.

Overall, the clustering findings reveal both surprising and expected findings; plate discipline does not necessarily equate to fewer strikeouts, strikeout rates do not impact wOBA very much, and selective batters are distinctly the most productive. This last point has limitations, however. The sample size for Cluster 1 is the smallest of the three clusters formed, and the players who fall under this category may be used differently from others. This can be seen from varying positions in a batting order, pinch-hit opportunities, and roster constructions. Further and deeper analysis is required to validate the preliminary trend found. Most importantly of these points, however, is that players can relate to any cluster and still perform well or poorly.

IX. Timeline

The last major step remaining in the project is to apply the second, and last, data mining technique to the dataset.

A. Anomaly Detection

Anomaly detection, as described earlier, will be implemented through the distance between data points and cluster centroid calculations and Z-Scores on wOBA and xwOBA differences. These separate algorithm executions will locate distinct anomalies of different kinds, deepening understanding of both player plate approaches and their subsequent performance results. The first half of this mining will be conducted from Mar. 16-20, and the second half from Mar. 23-27. Afterward, from Mar. 30-Apr. 2, the report for this phase will be developed and submitted, along with the proper documentation and organization needed in GitHub.

B. Final Deliverable

The final deliverable will include the final project report, featuring the findings and visualizations of the data. Additionally, the eProfile containing the project files will be organized and completed.

X. Anticipated Challenges

This last main stage of the project includes a few anticipated challenges. Firstly, it is crucial to select the best arbitrary threshold for locating anomalies during anomaly detection since there is no natural cutoff point. It cannot be too strict or too flexible, which suggests a range of somewhere between 5-10% of the farthest data points to be chosen as anomalies. Cluster size must also be taken into account, since each one has substantial differences in sample size. Anomaly detection through Z-Score computations faces some challenges as well. Differences may be small, but if the variance is low, then they will be calculated as larger Z-Scores, inflating the final result. These techniques will identify both structural and performance anomalies of players that deviate from the modeled structure reflected by the clusters.

XI. Conclusion

Exploratory data analysis, feature transformation, and clustering have laid out useful patterns in individual MLB batter data that are important to current players. Players share separate and unique plate approaches, varying in performance outcomes, across contact-based, swing-based, and discipline-based strategies. Discipline-based batters hold the greatest average offensive value in their performance outcomes. Anomaly detection will come next to identify exceptions to these groupings that branch from what is expected from them. This information can be put to use across MLB, such as for fans competing in fantasy baseball, teams deciding which players best suit their team, coaches putting together a batting order, or anything similar.

References

[1] Major League Baseball (MLB), "Baseball Savant," https://baseballsavant.mlb.com/, accessed Jan. 2026.

[2] M. J. Zaki and W. Meira Jr., *Data Mining and Machine Learning: Fundamental Concepts and Algorithms*. Cambridge, UK: Cambridge University Press, 2020.